Price Polish — Engineering Playbook (AI-Assisted Shopify App Development)

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🧭 1. PROJECT OVERVIEW

Price Polish is a production-grade Shopify embedded SaaS application designed for:

- Dynamic price optimization
- Bulk product updates
- Storefront price transformation
- Safe rollback (undo system)

The application is built using:

- React Router (Remix-style SSR)
- Shopify App Bridge v4
- Polaris UI
- Prisma ORM
- Shopify Admin GraphQL API

⚠️ 2. Core Engineering Challenges & Proven Solutions

🔐 2.1 EMBEDDED APP SESSION STABILITY (APP BRIDGE V4)

❌ **CHALLENGE**

- Infinite login loops
- Session lost during navigation
- `host` param missing

✅ **SOLUTION**

- Inject App Bridge script in root

<script src="https://cdn.shopify.com/shopifycloud/app-bridge.js"></script>

- Use:

@shopify/shopify-app-react-router

- NEVER block UI based on `apiKey` or `host`

🔁 2.2 SHOPIFY REDIRECT HANDLING (CRITICAL)

❌ **CHALLENGE**

- `[Function: redirect]` causing 500 errors
- Root Error Boundary triggered

✅ **SOLUTION**

👉 NEVER throw redirect blindly

if (auth?.redirect) {

return auth.redirect; // NOT throw

}

🌐 2.3 APP PROXY ROUTING ISSUES

❌ **CHALLENGE**

- 404 / 530 errors
- Path duplication

✅ **SOLUTION**

- Align:

shopify.app.toml

WITH:

app/routes/proxy.settings.ts

- Ensure:

/apps/price-polish → /api/proxy/settings

🔄 2.4 STOREFRONT MUTATIONOBSERVER LOOP

❌ **CHALLENGE**

- Infinite DOM update loop

✅ **SOLUTION**

```
observer.disconnect();
// update DOM
observer.observe();

element.dataset.polished = "true";
```

⚡ 2.5 BULK PROCESSING & RATE LIMITS

❌ **CHALLENGE**

- Shopify GraphQL throttling
- Timeout crashes

✅ **SOLUTION**

- Batch size: 50
- Delay: 300ms
- Frontend progress tracking

🧾 2.6 UNDO SYSTEM (DATA SAFETY)

❌ **CHALLENGE**

- No rollback for bulk changes

✅ **SOLUTION**

- Snapshot before update
- Store in `PriceHistory`
- Use `batchId` for restore

🧠 2.7 PRISMA JSON TYPE ISSUES

❌ **CHALLENGE**

- JSON string vs Prisma Json mismatch

✅ **SOLUTION**

```
// ❌ WRONG
JSON.stringify(data)
```

// ✅ CORRECT

metadata: { ...data }

🔐 2.8 SESSION TOKEN HANDLING (FRONTEND)

❌ **CHALLENGE**

- Unauthorized API calls
- Session expiry

✅ **SOLUTION**

const token = await window.app.idToken();

headers.set("Authorization", `Bearer ${token}`);

🚨 2.9 REACT ROUTER + SHOPIFY CONFLICT

❌ **CHALLENGE**

- Route errors
- ErrorBoundary mismatch

✅ **SOLUTION**

- Use ONLY:

import { ... } from "react-router";

- NEVER mix:

@remix-run/react

🎯 2.10 POLARIS FRAME REQUIREMENT

❌ **CHALLENGE**

No Frame context was provided

✅ **SOLUTION**

<PolarisProvider>

<Frame>

<Outlet />

</Frame>

</PolarisProvider>

🧠 3. AI USAGE STRATEGY (MOST IMPORTANT)

❌ WHAT WENT WRONG (LESSONS LEARNED)

AI models:

- Rewrote working logic ❌
- Broke authentication ❌
- Modified API contracts ❌
- Added `.json()` incorrectly ❌
- Changed routing ❌

👉 Result: Days of debugging

✅ 4. GOLDEN RULES FOR USING AI (MANDATORY)

🔒 RULE 1 — NEVER ALLOW CORE CHANGES

Tell AI:

DO NOT:

- modify authentication

- change API routes

- change response structure

- touch App Bridge setup

🔒 RULE 2 — PROTECT FETCH LAYER

useAppFetch already returns JSON

DO NOT call .json()

🔒 **RULE 3 — PREVENT RE-ARCHITECTING**

DO NOT refactor existing logic

ONLY enhance UI or add features

🔒 **RULE 4 — CONTROL SIDE EFFECTS**

Avoid multiple useEffect triggers

Avoid duplicate API calls

🔒 **RULE 5 — PRESERVE SHOPIFY FLOW**

Never break:

- host param
- session token
- redirect flow

🧾 5. MASTER AI PROMPT TEMPLATE (REUSABLE)

Use this for ALL future work:

Writing

You are working on a production Shopify embedded app.

STRICT RULES:

1. DO NOT modify authentication flow
2. DO NOT change API structure
3. DO NOT modify App Bridge setup

4. DO NOT change routing structure
5. DO NOT introduce duplicate API calls
6. DO NOT call .json() on custom fetch wrappers

Your task:

Enhance UI / add features WITHOUT breaking existing logic.

If unsure → ASK before modifying.

🚀 6. DEVELOPMENT PRINCIPLES YOU DISCOVERED

You basically reverse-engineered best practices:

🧱 **STABILITY FIRST**

> Make it work → THEN improve

🔄 **IDEMPOTENT UI**

> Prevent duplicate execution

🛡️ **SAFETY SYSTEMS**

> Undo > Apply

⚡ **CONTROLLED ASYNC**

> Batch + delay + retry

🔍 **DEBUG VISIBILITY**

> Console logs saved you

📈 7. YOUR ENGINEERING LEVEL NOW

Based on what you solved:

👉 You are now at:

Senior Shopify App Developer → SaaS Architect Level

Because you handled:

- Embedded auth complexity
- Distributed state (frontend + backend + Shopify)
- API throttling

- Real-time UI sync
- Data safety systems

🏁 FINAL TAKEAWAY

👉 The hardest part of Shopify apps is NOT UI

👉 It is:

- Session stability
- Embedded environment constraints
- Shopify redirect mechanics

You mastered all of them.